

Problem Library

1. **calculus/derivatives/chain-rule-01** *Chain rule with trig*

topics	calculus, derivatives	tags	chain rule, trig
difficulty	2	time	5
author	John Smith	created	2025-08-14
source	original		

date	course	term	type	assignment
2026-10-12	MATH 221	Fall 2026	exam	Exam 1
2026-09-21	MATH 221	Fall 2026	worksheet	Worksheet 3

Differentiate each function.

(a) (4 points) $f(x) = \sin(x^2)$

(b) (6 points) $g(x) = \cos^3(2x)$

2. **calculus/derivatives/product-rule-01** *Product rule*

topics	calculus, derivatives	tags	product rule
difficulty	1	time	4
author	John Smith	created	2025-08-14

date	course	term	type	assignment
2026-09-21	MATH 221	Fall 2026	worksheet	Worksheet 3

Let $h(x) = x^2 e^x$.

(a) (3 points) Find $h'(x)$.

(b) (3 points) Find the equation of the tangent line at $x = 1$.

(c) (4 points) Find all critical points of h .

3. (8 points) **calculus/integrals/u-sub-01** *Substitution*

topics	calculus, integrals	tags	u-substitution
difficulty	2	points	8
created	2025-09-02		

date	course	term	type	assignment
2026-10-12	MATH 221	Fall 2026	exam	Exam 1
2026-09-21	MATH 221	Fall 2026	worksheet	Worksheet 3

Evaluate $\int x \sqrt{x^2 + 1} dx$.

4. (6 points) **calculus/limits/squeeze-01** *Squeeze theorem*

topics	calculus, limits	tags	squeeze theorem, trig
difficulty	3	points	6
time	5	author	John Smith
created	2025-08-20		

date	course	term	type	assignment
2026-10-12	MATH 221	Fall 2026	exam	Exam 1

Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \cos(1/x)$.

5. **linear-algebra/eigen-01** *Eigenvalues of a 2x2 matrix*

topics	linear algebra	tags	eigenvalues, diagonalization
difficulty	3	time	10
author	John Smith	created	2026-01-10

date	course	term	type	assignment
2026-09-21	MATH 221	Fall 2026	worksheet	Worksheet 3

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

- (a) Find the eigenvalues and eigenvectors of A .
- i. (3 points) Eigenvalues.
 - ii. (3 points) Eigenvectors.
- (b) (4 points) Diagonalize A .